

Executive Summary: Physics Infrastructure for the Asymmetric Microfluidic Loop

Physics-Embedded Transformers Project

July 22, 2026

Abstract

This document summarizes the physics-only portion of the project: device geometry, region labeling, droplet occupancy, branch hydraulics, full-device CFD, flow-split calibration, and interpolation of the computed continuous-phase velocity field onto measured droplet trajectories. It intentionally avoids machine-learning architecture and training details. The present physics pipeline produces a geometrically clean full-device computational domain, a validated finite-element Stokes solution on the frozen $24\text{ }\mu\text{m}$ production mesh, a calibrated library of flow splits from 0.40 to 0.64 left fraction, and CFD-enriched droplet tracking data for video 2.

1 Physical System

The device is an asymmetric loop microfluidic channel. A single inlet channel feeds an upstream junction, where the flow separates into a long left branch and a shorter right branch. The two branches reconnect at a downstream junction and discharge through a single outlet channel. The channel width and height are both $100\text{ }\mu\text{m}$, corresponding to 25 pixels at the calibrated $4\text{ }\mu\text{m}/\text{px}$ image scale.

The measured branch lengths are:

$$L_{\text{left}} = 1791\text{ }\mu\text{m}, \quad L_{\text{right}} = 1491\text{ }\mu\text{m}.$$

The resulting geometric asymmetry is:

$$\frac{L_{\text{left}}}{L_{\text{right}}} = 1.201207, \quad \alpha_{\text{geom}} = \frac{L_{\text{left}} - L_{\text{right}}}{L_{\text{right}}} = 0.201207.$$

The working physical picture is that droplets occupy finite portions of the network and alter the effective branch resistance. The continuous phase is treated as a low-Reynolds-number background flow field, while droplet occupancy and branch hydraulics provide framewise physical state variables.

2 Physics Tools

The physics pipeline is organized by scientific responsibility rather than by implementation history:

1. **Geometry and region labeling:** reconstruct the calibrated device coordinate system, centerlines, branches, junctions, and channel-region labels.
2. **Droplet occupancy:** compute fractional droplet occupancy in physical regions of the device.
3. **Branch hydraulics:** estimate framewise branch resistance and flow split from normalized branch occupancy.

4. **Full-device CFD:** solve the steady continuous-phase Stokes problem over the full loop geometry.
5. **CFD library and interpolation:** calibrate a library of flow splits and sample the corresponding velocity field at arbitrary valid points in the device.
6. **Physics-enriched tracking:** append local CFD velocity and hydraulic state to measured droplet centroids.

These tools are deterministic preprocessing and simulation components. Their outputs are reusable physical data products.

3 Geometry and Mesh

The full device is constructed as a continuous CAD-style fluid region using the known centerline topology and the fixed $100\ \mu\text{m}$ channel width. The domain is represented in physical units, with coordinates stored in micrometers. The final computational domain has one connected fluid component and one central island. Dedicated topology checks confirmed no self intersections, no dangling boundaries, no duplicated interior wall fragments, and no unintended internal boundaries.

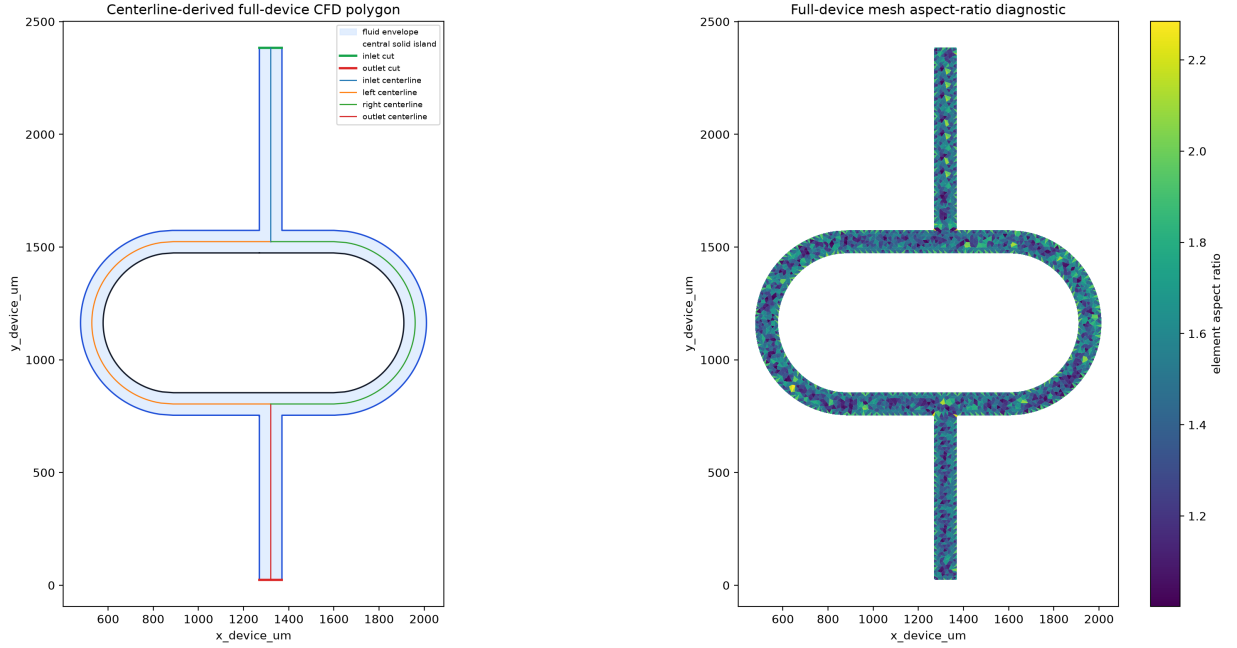


Figure 1: Left: full-device CAD polygon with centerlines. Right: mesh-quality visualization for the frozen production mesh.

The frozen production mesh uses the $24\ \mu\text{m}$ configuration. Mesh quality is summarized in Table 1. The mesh is fine enough to resolve the junctions and curved walls while avoiding unnecessary refinement in the long straight sections.

Table 1: Frozen 24 μm production mesh quality.

Quantity	Value
Nodes	2339
Triangular elements	3807
Minimum angle	25.73°
Maximum angle	122.39°
Median aspect ratio	1.456
Maximum aspect ratio	2.285
Fluid area	$5.226 \times 10^5 \mu\text{m}^2$
Boundary length	$1.043 \times 10^4 \mu\text{m}$

4 Continuous-Phase CFD

The continuous phase is modeled as a steady, incompressible, creeping flow. The governing equations are the Stokes equations:

$$-\nabla p + \mu \nabla^2 \mathbf{u} = \mathbf{0}, \quad \nabla \cdot \mathbf{u} = 0,$$

where $\mathbf{u}$ is the two-dimensional velocity field, p is pressure, and μ is the dynamic viscosity of the continuous phase. No-slip conditions are imposed on solid walls. The inlet carries a fixed incoming flux. Outlet and branch-flow conditions are formulated so that calibrated left/right flow fractions can be generated without changing the geometry.

The validated equal-split case corresponds to

$$f_L = \frac{Q_L}{Q_L + Q_R} \approx 0.50.$$

For this case, the left and right branch fluxes are equal to numerical precision, and the velocity and pressure fields are smooth over the full device. The maximum speed in the production-library cases is approximately 0.0817 m/s. Pressure ranges over the calibrated cases are about 173–190 Pa, increasing as the imposed branch bias is made stronger.

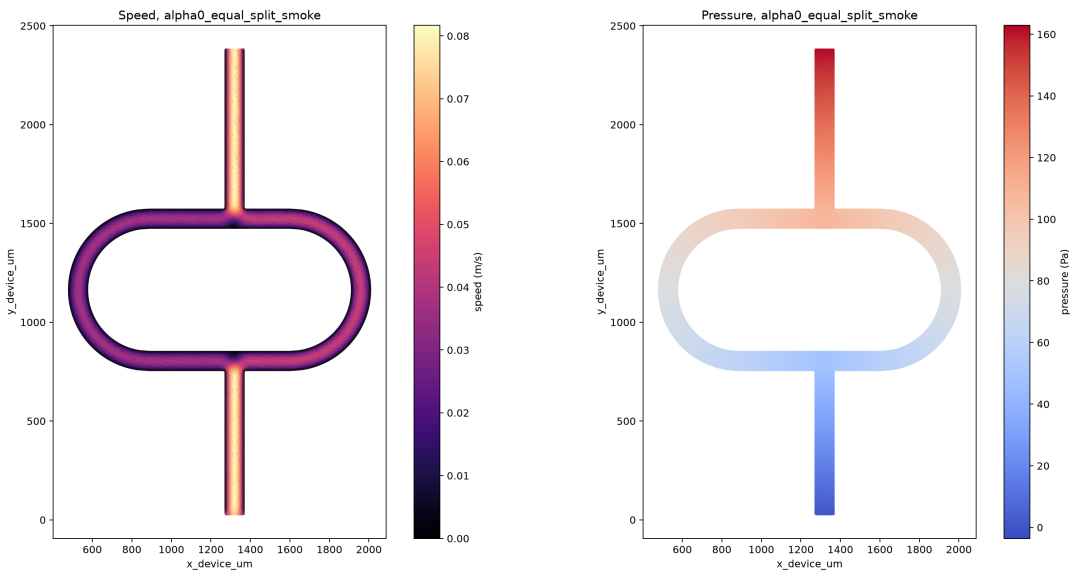


Figure 2: Representative equal-split full-device CFD solution: velocity magnitude and pressure field.

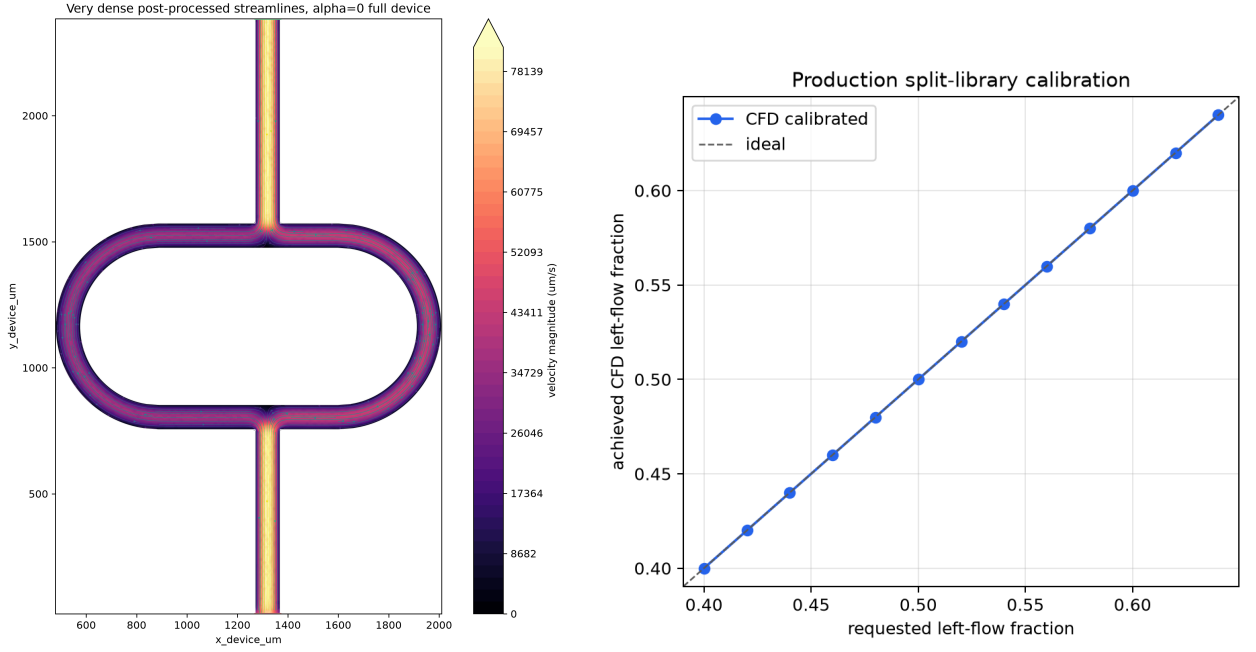


Figure 3: Left: dense streamline post-processing for the equal-split case. Right: production calibration curve for requested versus achieved left-flow fraction.

5 Flow-Split Calibration Library

The production CFD library spans requested left-flow fractions from 0.40 to 0.64 in increments of 0.02. The achieved split is used as the interpolation coordinate. Calibration succeeded for all 13 requested targets, with no unreachable cases. The natural unmodified split is

$$f_{L,\text{natural}} = 0.458119.$$

The maximum absolute calibration error over the production library is

$$3.61 \times 10^{-9},$$

and the achieved splits are strictly monotonic. The largest adjacent gap is 0.020000004, matching the intended resolution.

Table 2: Production CFD split-library summary.

Quantity	Value	Interpretation
Successful cases	13	All requested targets reached
Requested range	0.40–0.64	Left-flow fraction
Natural split	0.458119	Unbiased full-device flow split
Maximum calibration error	3.61×10^{-9}	Achieved vs. requested split
Largest adjacent gap	0.020000004	Library spacing
Monotonic achieved splits	yes	Suitable for interpolation

6 Mesh and Field Verification

A mesh convergence comparison was performed between the frozen 24 μm production mesh and a finer 12 μm mesh. The goal was not to replace the production mesh, but to verify that the velocity field, pressure field,

and flux structure are stable at the selected resolution. The largest differences occur near the inlet and outlet junctions, which is expected because these are the regions with the strongest geometric curvature and flow redistribution. The production mesh was retained as the frozen default after this verification: it captured the same qualitative field structure as the finer mesh while remaining inexpensive enough for production library generation.

7 Physics-Enriched Droplet Data

The full-device CFD library is sampled at measured droplet centroids using the framewise hydraulic split as the library coordinate. This attaches local continuous-phase velocity information to each observed droplet without extrapolating outside the CFD fluid domain.

For video 2, the enriched tracking table contains 432,460 observed droplet rows. Of these, 429,929 rows are inside the valid CFD fluid domain:

$$\text{valid fraction} = 0.994147.$$

The remaining invalid rows are confined to the outlet channel and are consistent with the known downstream video crop/metadata exclusion rather than a CFD failure.

Table 3: Regional validity of CFD samples on measured droplet centroids.

Region	Rows	Invalid rows	Invalid percentage
Inlet channel	65,614	0	0.00%
Inlet junction	26,553	0	0.00%
Left branch	100,083	0	0.00%
Right branch	170,729	0	0.00%
Outlet junction	8,399	0	0.00%
Outlet channel	61,082	2,531	4.14%

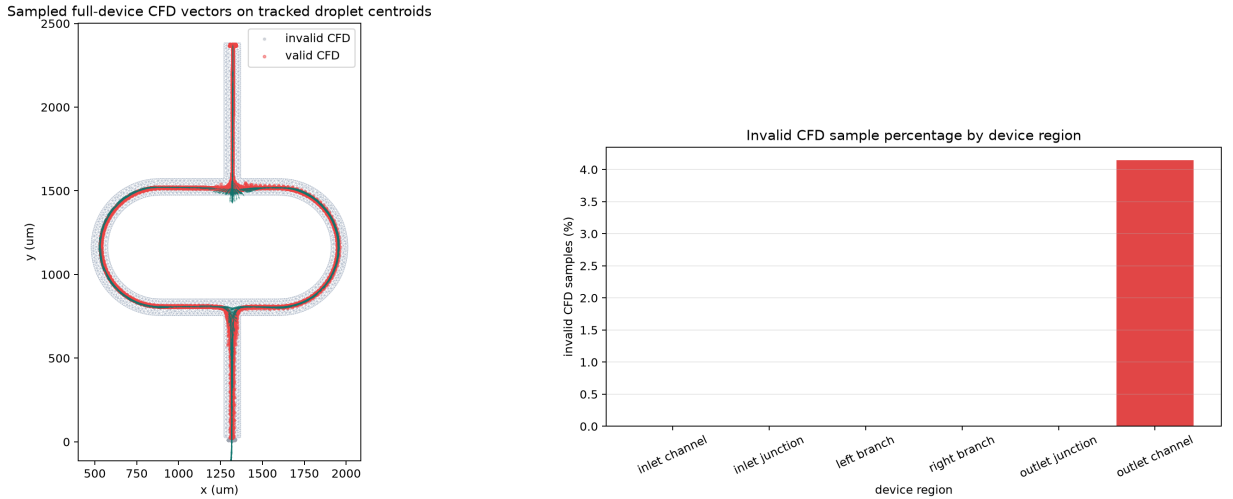


Figure 4: Left: sampled CFD vectors over tracked droplet centroids. Right: invalid CFD percentage by physical region. Invalid samples are isolated to the cropped outlet-channel region.

8 Current Physics Status

The physics infrastructure is in a production-ready state for the current single-device, single-experiment analysis:

- the full-device geometry is topologically clean;
- the 24 μm mesh is frozen as the production mesh;
- the full-device Stokes solve produces smooth pressure and velocity fields;
- the calibrated CFD library provides monotonic flow splits from 0.40 to 0.64;
- the library is interpolation-ready using achieved left-flow fraction;
- measured droplet tracks are enriched with valid local CFD velocity for 99.4% of observed rows.

Future physics extensions can build on this foundation by adding explicit droplet-induced resistance laws, pressure-loss models at the junctions, time-dependent coupling, or additional experimental devices. These are natural physics-modeling extensions and do not require revisiting the validated geometry, mesh, and production-library foundation summarized here.